Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

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***Abstract*—Hybrid retrieval over surveillance video combines a structured metadata filter with a dense semantic index under a *fixed* mixing weight, degrading accuracy on counting and action queries. Three contributions raise mean Precision@5 (macro-averaged over five categories) on a 150-query benchmark of the four-camera deployment: 0.67 (best fixed-α, α=0.3) to 0.83. (C1) An *intent-adaptive blending function* $\alpha(F, \iota, n_s, n_v)$ gives a closed-form per-query weight from filter F, intent ι and branch-size imbalance, plus hard overrides for counting and action queries (Eq. (3)); fixed-α schemes cannot match all per-category optima (observed empirically). (C2) A *heterogeneous-space person descriptor* fuses a SigLIP visual embedding ($\mathbb{R}^{768}$) with a MiniLM attribute-text embedding ($\mathbb{R}^{384}$) into one L2-normalized $\mathbb{R}^{1152}$ vector by weighted concatenation, not same-space addition; an ablation shows a 10.7-pp F_1 gain over the best unimodal index and 2.2-pp over rank-based late fusion (RRF). (C3) A *conversational alert grammar* materializes fully-typed alert rules (event, object, zone, threshold, time, severity, cooldown, side-effects, cadence) from free text via a released keyword/regex materializer (defaults scored when unspecified), reducing manual rule authoring to a clarification round-trip; we report 91% rule-correctness in a separate 60-utterance deployment study (LLM-backed materializer). They run in a nine-model real-time pipeline sustaining 2 FPS per camera over four cameras, with a 109 ms median FAISS lookup over 312k person descriptors. The supplementary artifact exercises the code paths behind the deployment numbers end-to-end; the deployment per-query export was not preserved, so its CIs and p-values are not re-computable.**

***Index Terms*—Hybrid Retrieval, Intent-Adaptive Ranking, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Conversational Alert Authoring, Vision–Language Surveillance.**

I. Introduction

NATURAL-language interfaces over surveillance video are feasible: instance segmentation, person Re-ID and contrastive vision–language models are mature, and structured-output LLMs turn an utterance into a typed query schema. The unsolved problem is *how to rank* candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) give two incommensurable scales. Dominant fusion strategies: a fixed convex sum $\alpha S_{\text{struct}} + (1-\alpha) S_{\text{sem}}$ tuned once per system [1]–[3], parameter-free Reciprocal Rank Fusion [4], or sequential filter-then-rank pipelines [5]. Empirically (Table I) no single α serves counting, finding, tracking and action queries simultaneously: $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and $\alpha = 1.0$ to 0.04 P@5 on find queries, and no fixed value exceeds 0.67 macro-mean P@5.

Two further problems compound this. Prior multimodal Re-ID work [6], [7] either fuses same-space embeddings by addition or trains a joint dual encoder, both needing dimension-matched backbones or joint contrastive data that deployments rarely have. Form-driven alert engines need an IT operator to translate a behavioral description into a typed rule, capping useful rules.

Contributions

1) **Intent-adaptive hybrid retrieval** (§IV). Closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap observation (Observation 1), stated empirically; +0.16 macro-mean P@5 over the best fixed-α baseline.
2) **Heterogeneous-space person fusion** (§V). Weighted-concatenation descriptor fusing backbones of incompatible dimensionality without joint training; +10.7-pp F_1@10 over the strongest unimodal index, +2.2-pp over rank-based fusion (RRF).
3) **Conversational alert authoring** (§VI). Typed 9-field grammar with keyword/regex materializer (defaults scored when unspecified); 91% full-rule correctness on a 60-utterance held-out deployment study.
4) **An end-to-end open implementation** on a 4-camera live deployment whose data are not released, plus a released single-camera reproducibility track exercising

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the C1–C3 code paths (§VII).[1]

II. Related Work and Position

We compare with the closest published work along the three axes below; only this one provides all three, and none combines per-query adaptivity with heterogeneous-space Re-ID fusion.

Hybrid structured/semantic retrieval. Dense plus structured retrieval is established IR practice [3], [4], [8], mainly via fixed-α convex combination or parameter-free RRF [4]; RAG [3] is retrieval-augmented generation, not a tuned blend. IR query-dependent weighting (learning-to-rank, query-dependent fusion) learns weights from large-scale click data on text-only sparse/dense pairs, lacking hard overrides for infeasible branches. HyST [5] (sequential LLM-filter then dense) and Agentic-RAG [9] (router-based) still produce no per-query blended score; surveillance forensic-search systems (VIDEX [1], three-stage DL [2]) use a single retrieval branch. Per-query mixing from parsed intent and result-set evidence with category hard overrides is undemonstrated for surveillance video to our knowledge.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders is an IR retrieval pattern; we evaluate it as the *primary* surveillance-Re-ID descriptor. Prior NL-based person-search [7] couples a dual encoder by joint training or descriptive fusion over same-dimensional modalities; rank-based late fusion [4] needs two indices and score normalization. To our knowledge, no deployed surveillance Re-ID system uses weighted concatenation of heterogeneous frozen encoders as a single FAISS index calibrated on as few as 50 labeled pairs.

Conversational alert authoring. Form-driven surveillance editors expose rule builders, and structured-output LLMs [10] are standard for slot-filling; the contribution is reusing the same query-parsing schema to materialize a 9-tuple typed alert rule end-to-end, not a new LLM capability.

III. Background and System Substrate

The nine-model real-time pipeline runs capture, perception, indexing, reasoning and output layers; all but C1–C3 are off-the-shelf. Live RTSP / MJPEG streams (1080p, 2 FPS) and uploads are processed by YOLOv11m-seg [11], BoT-SORT [12] with OSNet $x1_0$ [6] for MOT/Re-ID, OpenVINO PAR-crossroad-0238 for seven multi-label attributes, and a CIE LAB histogram with CIEDE2000 naming over a 25-class CSS3 palette. Two FAISS `IndexFlatIP` indices [8] hold a scene embedding ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [13]) and the person descriptor ($\mathbb{R}^{1152}$, §V). MongoDB stores 12 compound-indexed collections behind a FastAPI backend and Next.js dashboard. Captions come from a tiered VLM pipeline (Qwen2-VL-2B with ViT-GPT2 / zero-shot CLIP fallbacks); the NL parser is a structured-output LLM (Beam in the released track; the code accepts any OpenAI-compatible endpoint). Model cards for Qwen2-VL, PAR, CIEDE2000 and MMR are omitted for the 15-reference cap.

[1]Code, benchmark definitions and measured outputs are in the supplementary artifact; the public repository URL is withheld from this preprint and given on acceptance.

TABLE I
Fixed-α baselines, 150-query NL benchmark (30 per category); **Mean**: macro-average over five categories (bold = best, italic = worst).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	*0.55*	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (3))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

IV. C1: Intent-Adaptive Hybrid Retrieval

Problem: A natural-language query q is parsed into a structured filter F (Pydantic `NLFilter`, 14 typed fields, incl. `action`) and a semantic payload q_{sem}. Structured branch: $R_s = \{r\}$ from MongoDB with $S_{\text{struct}}(r) \in [0,1]$ (duration max-scaled per query); semantic branch: R_v from FAISS with per-query min-max normalized $S_{\text{sem}}(r) \in [0,1]$ (`score_norm`). The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0,1]. \tag{1}$$

Failure of fixed-α schemes: We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table I; α=0.0 proxies the 0.10 semantic-dominant override). *Counting* queries (*"how many cars in Zone A"*) need an exact integer only the structured branch produces: a continuous semantic score cannot express "equals 5," so semantic weight dilutes a correct count (P@5 0.94 at α=1.0 $\rightarrow$ 0.31 at α=0.0). Counting top-5 are the highest-scoring supporting clips, relevant iff they contain the true count; S_{struct} is the duration-normalized count-constrained segment score per record; exact-count EM/MAE is complementary, in the release. *Action* queries (*"person running"*) carry an `action` field and are routed semantic-only, so the α=1.0 column of that table must not be read as structured-branch noise injection: the α-sweep harness instead scores them by a time-overlap proxy, and P@5 falls 0.69 (α=0.0) $\rightarrow$ 0.00 (α=1.0). Best fixed macro-mean 0.666 at α=0.3, motivating a per-query not per-deployment weight. The release gives one success and one failure per regime, including an intent-routing error.

Adaptive blending function: Let $C : \mathcal{F} \rightarrow \{\text{NUM}, \text{ACTV}, \text{SOFT}\}$ be the *category function* on parsed filters:

$$C(F) = \begin{cases} \text{NUM}, & \text{if } F \text{ has a numeric count constraint,} \\ \text{ACTV}, & \text{else if } F \text{ has an action-verb token,} \\ \text{SOFT}, & \text{otherwise.} \end{cases} \tag{2}$$

We define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F)=\text{NUM}, \\ 0.10, & C(F)=\text{ACTV}, \\ \text{clip}_{[0.10,0.95]}(\alpha_{\text{soft}}), & \text{else}, \end{cases} \tag{3}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{4}$$

where NUM (numeric-count override) and ACTV (action-verb override) are $C(F)$'s two override categories (Eq. (2)), distinct from soft intent labels ι (substring match). Intent prior $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ respectively (0.7 when unmatched); complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$ over the non-empty fields among six keys (camera id, object name, object colour, zone, location hint, timestamp); evidence ratio $\rho_{\text{src}} = n_s / \max(1, n_s + n_v)$, $n_s = |R_s|$, $n_v = |R_v|$. The overrides encode domain truths: a count constraint is unanswerable from semantic similarity, an action verb is routed semantic-only.

Why fixed α cannot match: Observation 1 (override gap, empirical). No single fixed α simultaneously attains the count optimum at α=1.0 (0.94) and the action optimum at α=0.0 (0.69) on our 150-query benchmark; the optima are 0.9 apart, and adaptive-α attains the per-category maxima empirically on this benchmark. This is an empirical observation over $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, not a theorem: we assume no concavity and claim no general lower bound.

Soft-zone behavior: For non-override queries, Eq. (4) interpolates intent prior, complexity, and per-branch evidence (weights $0.6/0.2/0.2$, prioritizing the parsed intent prior; ρ_{cplx} and ρ_{src} are bounded corrections). The 16-point soft-weight sweep in §VII shows the result does not depend on the exact soft weights. Realized α: 1.00 (count), 0.10 (action); the released control's soft zones realize 0.60–0.66 (find), 0.62 (track), 0.65–0.74 (filter), inside the $[0.10, 0.95]$ clip.

Post-ranking: Top-k candidates are re-ranked by a recency boost $S' = S_{\text{hyb}} \cdot (1 + 0.5 \cdot 2^{-age/24\text{h}})$, age in hours (range $[1.0, 1.5]$, 24-h half-life), and MMR (λ=0.3).

V. C2: Heterogeneous-Space Person Fusion

Problem: Multimodal person Re-ID [7] typically fuses two same-dimensional encoders; a deployment must use the pre-trained encoders at hand—here SigLIP-base ($\mathbb{R}^{768}$) [14] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [15], of differing dimensions.

Construction: Per person ROI we form

$$\begin{aligned} \mathbf{v} &= \text{SigLIP}(\text{Mask}(\text{ROI})) \in \mathbb{R}^{768},\ \text{L2-normalized} \\ \mathbf{t} &= \text{MiniLM}\big(\text{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384},\ \text{L2-normalized}, \end{aligned} \tag{5}$$

where $\text{Text}(\hat{y}, C)$ is the sentence from the seven sigmoid predictions $\hat{y}$ and the color C (e.g., *"person wearing red clothing, carrying bag, long sleeves, male"*). Table II uses paired image+attribute-text queries, so (α_v, β_t) matter. Text-only person search is a fallback ($\mathbf{v}$=$\mathbf{0}_{768}$, $\mathbf{t}$=MiniLM(q)) whose ranking equals text-only cosine, so the fusion gains do not come from it. Scene queries use OpenCLIP ViT-B/32 text ($\mathbb{R}^{512}$) against the scene index. Person gallery: N=312,544 descriptors from the 8-h deployment ($1152 \times 4\,\text{B} \approx 1.44\,\text{GB}$ flat storage before FAISS overhead); calibration uses a held-out 50-pair split (no CV; gallery disjointness not logged).

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \tag{6}$$

TABLE II
PERSON-INDEX ABLATION. "ADD (PROJ.)" ADDS $\mathbf{t}$ PROJECTED TO $\mathbb{R}^{768}$ BY A SINGLE-SEED RANDOM GAUSSIAN PROJECTION (ILLUSTRATIVE); ZERO-PAD/NORM-MATCHED ADDITION WOULD BE FAIRER. "LATE (RRF)" IS RRF [4] ON TWO INDICES; "CONCAT (OURS)" IS EQ. (6).

Person index	P@5	R@10	F_1@10
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v} + \mathbf{P}\mathbf{t}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ from a coarse grid search on the unit simplex (step 0.1, nine pairs) maximizing F_1@10 on the held-out calibration set, where F_1@10 $\triangleq$ H-mean(P@5,R@10); the standard P@10-based F_1 is lower. The ablation sweep varies gallery-side weights on paired queries, so the $(0.5, 0.5)$ vs. RRF tie and $(0.4, 0.6)$–$(0.7, 0.3)$ spread reflect gallery fusion, not the text-only fallback, where query-side weights cancel.

Why concatenation, not addition: Same-space addition $\mathbf{v} + \mathbf{t}$ needs a projection $\mathbf{P} \in \mathbb{R}^{768\times384}$ and is norm-sensitive; weighted concatenation preserves both encoders' full rank, needs no joint training, and learns only the scalar pair (α_v, β_t). Our contribution is the heterogeneous concatenation as the primary single-index Re-ID descriptor, not concatenation itself, established in IR.

Ablation: The concat construction beats the strongest unimodal baseline by 10.7 pp F_1@10 and the strongest fusion baseline (RRF) by 2.2 pp, using one FAISS index instead of two. The coarse (α_v, β_t) sweep is unimodal around $(0.6, 0.4)$ (the $(0.5, 0.5)$=0.785 tie with RRF needs multi-seed re-verification).

Indexing complexity: Adding a person is $O(d) = O(1152)$; querying k neighbors over N persons with `IndexFlatIP` is $O(Nd)$. On our hardware the person-index median is 109 ms for N = 312,544 descriptors, a separate deployment measurement.

VI. C3: Conversational Alert Authoring

Problem: Commercial alert engines [1], [2] require an operator to turn a behavioral intention into a typed rule—a productivity bottleneck.

Grammar: A typed alert rule is the 9-tuple $\mathcal{R} = \langle\, e, \mathcal{O}, \zeta, \tau, \mathcal{T}, s, c, \Sigma, \kappa \,\rangle$ with $e \in \{\text{crowd}, \text{loiter}, \text{fight}, \text{zone}, \text{count}, \text{time}\}$, $\mathcal{O} \subseteq \mathcal{C}_{\text{det}}$ (detected classes), ζ (zone), τ (numeric threshold: operator/value/unit), $\mathcal{T} = [t_1, t_2]$ (time window, cross-midnight), $s \in \{\text{info}, \text{warning}, \text{high}, \text{critical}\}$ (severities, default `info`), $c \in \mathbb{N}$ (cooldown seconds, default 60), $\Sigma \subseteq \{\text{store_clip}, \text{snapshot}, \text{push_ws}\}$ (fixed side-effects), and $\kappa \in \mathbb{R}_{>0}$ (evaluation cadence, Hz, default 1Hz). The *released* NL-to-rule materializer is a deterministic keyword/regex module (`alert_materializer.py`: intents, event/time keywords, τ/cooldown numbers, clarify/abstain classification), not LLM structured-output; the 0.91 rule-correctness of §VII came from the deployment study with the LLM-backed materializer and is not reproducible from the released benchmark (Limitations).

Authoring example: The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces

$$\mathcal{R} = \langle \text{crowd},\ \{\text{person}\},\ \text{Zone A},\ (>,5),\ [22{:}00, 06{:}00],\ \text{info},\ 60,\ \{\text{snapshot}, \text{push_ws}\},\ 1\text{Hz} \rangle$$

without manual configuration: severity, cooldown, side-effects and cadence default when unstated, and $[22{:}00, 06{:}00]$ applies the default 06:00 end when none is specified (night $\rightarrow$ 20:00–06:00 policy). Six rule families run per frame—crowd ($\#\{\text{persons}\} \geq \tau_{\text{crowd}}$ within camera/zone), loitering ($\Delta t_{\text{track}} \geq 60$ s for any track), fight ($\bar{\mathcal{E}}_W \geq 0.03$ with two person centres < 24 px apart, overlapping persons at 1080p), zone occupancy (persons in ROI $>$ capacity), count ($\#\{c\} \square n$, $\square \in \{=, \geq, \leq, >, <\}$) and time-of-day ($t \in [t_{\text{start}}, t_{\text{end}}]$, cross-midnight)—each cached from MongoDB with a 5-second TTL. Ambiguous cases (fuzzy zone, implicit "a few", cross-midnight) account for the 9% non-perfect rules in §VII.

Validation safeguards (clarify/abstain): Unknown zones, missing thresholds or time windows, and unsupported verbs (e.g., "follow him home", "identify the person") trigger schema validation requesting clarification, not a rule. Stress-kind classification and 9-tuple validation are deterministic and tested (`backend/tests/test_alerts.py`).

Complementary statistical anomaly detector: To surface unknown behaviors we keep a rolling 7-day baseline of detection counts $\{x_{c,h}^{(d)}\}_{d=1}^{7}$ per (camera, hour) bucket and emit an event when

$$Z_{c,h} = \frac{x_{c,h}^{*} - \mu_{c,h}}{\sigma_{c,h} + \varepsilon} \geq \tau_Z, \tag{7}$$

with $\tau_Z \in \{1.5, 2.0, 3.0\}$ mapping to $\{\text{low}, \text{medium}, \text{high}\}$ for unusual-object, off-hours and crowd-spike events; $+\varepsilon$ guards $\sigma{=}0$. Unusual-object uses a count heuristic, off-hours fires when no baseline exists, and crowd-spike uses $z{\geq}2.0$. On a 7-day labeled subset, precision/recall/F_1 reach $0.872/0.811/0.840$ at 1.4 FP/day (crowd-spike), $0.804/0.866/0.834$ at 2.1 (off-hours) and $0.713/0.792/0.751$ at 3.7 (unusual-object, acknowledged high).

The detection-and-indexing loop is off-the-shelf: per frame, YOLOv11m-seg [11] produces detections, each person box is cropped and letterboxed for the PAR attribute head [11] and LAB-histogram colour naming, its text attribute and colour feed the person descriptor of §V, and the record goes to MongoDB.

Query resolution. The NL query q is parsed by an LLM (Beam in the released track; the released code accepts any OpenAI-compatible endpoint) into a filter F, a semantic query q_{sem} and an intent ι; candidates come from $R_s \leftarrow \text{MongoDB.find}(F)$ and $R_v \leftarrow \text{FAISS.search}(g_T(q_{\text{sem}}), k)$, with g_T OpenCLIP text for scene queries and MiniLM ($\mathbf{v}{=}\mathbf{0}$) for person queries against the 1152-d index. The weight is $\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ (Eq. (3); action queries skip R_s, so α is inert in production); the branches merge and rescore by Eq. (1), then pass through a confidence filter, the recency boost and MMR of §IV, and the top-k clips plus an LLM answer are returned.

VII. Evaluation

Setup: Hardware: Intel i7-13700K, RTX 4070 (12 GB), 32 GB DDR5, Windows 11; Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. **Full deployment.** Four 1080p IP cameras at 2 FPS per stream for $\sim$8 h ($\sim$230k frames, 1.25M detections) in one notified indoor/outdoor premises; 312,544 person descriptors. **Deployment benchmark.** A 150-query NL benchmark (30 each: count, find, track, action, filter) authored from live clips, annotated for relevance; for counting, relevance means the clip contains the true count and P@5 measures supporting-clip retrieval. Its per-query export, annotator roster and tune/test partition were not preserved, so deployment CIs below are query-level and its numbers are recorded measurements. **Released track.** A single-camera subset (123 clips, 640×480, 5 FPS, 3.6 s each, $\sim$7.2 min in a 2 h window; 242 scene frames; 1,778 person descriptors) with metadata-derived rather than human labels, 2,016 track-disjoint person pairs (528 calibration); harness `eval_*.py`, outputs `results/`; the deployment study used a 50-pair calibration split. Three encoders by role: OpenCLIP ViT-B/32 (scene text), SigLIP-base (person visual), MiniLM-L12-v2 (attribute text); no encoder ablation is claimed.

Statistical protocol. Mean P@5 with 95% bootstrap CI (10,000 query-level resamples); paired comparisons (adaptive vs. fixed-α, concat vs. each fusion baseline) use a two-sided paired bootstrap on per-query P@5 deltas with a Wilcoxon cross-check, $p < 0.05$ without multiple-testing correction. Grouped resampling could not be recomputed for the deployment; the released track reports a grouped bootstrap over video/time groups with Holm-corrected per-category tests (3,000 resamples) in `backend/app/eval/stats.py`.

Evaluation protocol (definitions and thresholds): Precision@5 $= |\text{top}{-}5 \cap \text{relevant}|/5$; Recall@10 $= |\text{top}{-}10 \cap \text{relevant}|/|\text{relevant}|$; F_1@10 is the harmonic mean of P@5 and R@10 (not the standard P@10-based F_1). Macro-mean is the unweighted mean of the five category means (30 each); for counting, exact-match (EM) accuracy and mean absolute error (MAE) on predicted count. Thresholds follow the C3 rule families above and Eq. (7) (detector confidence 0.5; fight trigger 0.03 at $<$ 24 px; Z-score severities $1.5/2.0/3.0$). Deployment tables keep their recorded single-run measurements.

C1 – adaptive-α vs. fixed-α: Adaptive-α raises macro-mean P@5 from 0.666 (best fixed-α, $\alpha = 0.3$) to 0.828: gain $+0.162$, paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table III). A per-intent fixed-α lookup from Table I (count$\rightarrow$1.0, action$\rightarrow$0.0, filter$\rightarrow$1.0, find/track$\rightarrow$0.3) reaches 0.822 macro-mean, so soft terms ($\rho_{\text{cplx}}, \rho_{\text{src}}$, $0.6/0.2/0.2$) contribute $+0.006$; paired bootstrap and Wilcoxon signed-rank tests reject equality with the best single fixed-α ($p < 0.0002$, the 10,000-resample bootstrap's resolution floor); per-category gains over the best fixed-α are zero on count, action and filter (where $\alpha = 1.0$ saturates) and positive on find ($+0.02$, $p = 0.04$) and track ($+0.01$, $p = 0.07$), both marginal

TABLE III

DEPLOYMENT MEASUREMENTS; PER-QUERY EXPORT NOT PRESERVED, SO CIS AND p-VALUES ARE NOT RE-COMPUTABLE FROM THE RELEASE. PER-CATEGORY P@5 WITH 95% BOOTSTRAP CIS (10,000 RESAMPLES, n=30); Δ = MEAN OF (ADAPTIVE − BEST FIXED-α), p = TWO-SIDED PAIRED BOOTSTRAP (FLOOR 0.0002). LAST ROW: BEST SINGLE FIXED-α ($\alpha = 0.3$) VS. ADAPTIVE-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 ($\alpha = 1.0$)	0.94 ± 0.05	$+0.00$ (n.s.)
Find	0.76 ± 0.07 ($\alpha = 0.3$)	0.78 ± 0.07	$+0.02$ (0.04)
Track	0.80 ± 0.06 ($\alpha = 0.3$)	0.81 ± 0.06	$+0.01$ (0.07)
Action	0.69 ± 0.08 ($\alpha = 0.0$)	0.69 ± 0.08	$+0.00$ (n.s.)
Filter	0.92 ± 0.05 ($\alpha = 1.0$)	0.92 ± 0.05	$+0.00$ (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ (< 0.0002)

and not surviving Holm correction. Crucially, no single fixed α matches the count, action and filter optima simultaneously (Observation 1); adaptive-α does empirically.

Released-subset check (C1): The same code path was re-run on the released single-camera subset with a *post-hoc* protocol-matched control (50 tune / 100 test queries, metadata-derived labels; `eval_retrieval_control.py`). The adaptive rule reaches macro P@5 0.538, equal to the best tune-selected fixed α (0.538 at $\alpha = 0.7$; $\Delta = 0.000$, $p = 1.0$) and above the fixed per-intent lookup (0.428), while extremes collapse (0.048 at $\alpha = 0.0$, 0.380 at $\alpha = 0.3$). It leads where overrides fire (count 0.390 vs. 0.010 at $\alpha = 0.0$; action 0.090 vs. 0.000 at $\alpha = 1.0$), so overrides are not harmful; but best fixed α matches the adaptive rule on all five categories, so the subset does not reproduce the deployment gain of Table I. The release ships a *raw*, non-protocol-matched smoke benchmark (no target count in its count questions; an exploitable time window on its action queries) ranking adaptive below the best fixed α (0.486 vs. 0.568; oracle best 0.578) and preferring $\alpha = 1.0$ on action queries (0.520); the post-hoc control removes both mismatches and is the number we quote. *End-to-end.* With the deployed parser's answer (Beam) driving the structured branch on the identical protocol (`eval_end_to_end.py`), adaptive-α reaches macro P@5 0.450 vs. 0.448 for the best tune-selected fixed α ($\Delta = +0.002$, p=0.80, not significant); the parser answered all 150 queries (mean 3.0 s), giving the reference colour on $90/90$ colour-bearing queries and a numeric count target on $30/30$ counting queries. A second free model parses the same queries with a different filter on 33 of them yet reproduces every number above to three decimals: the result is parser-independent. The 0.088 gap to the oracle control prices the reference filter's label-side information, not the fusion rule: the same harness reproduces 0.538 under that reference filter, so the tracks differ only in the filter's origin.

Competitors and robustness (C1): Three further comparisons use the same control. (i) Learned competitor: a multinomial logistic model predicting the best fixed α per query from intent ι, branch sizes n_s, n_v and filter size $|F|$, fitted on the 50 tune queries, reaches 0.534 on the 100 test queries, below the training-free rule's 0.538: its oracle-best-α label is 0.5 on 49 of the 50 tune queries, so it collapses to the constant 0.5; it matches the closed-form rule on both override categories (count 0.39, action 0.09) and loses only in the soft zone (find 0.91 vs. 0.92, filter 0.47 vs. 0.48). (ii) Soft-weight sensitivity: all 16 triples of the soft weights of Eq. (4) summing to one, searched on the tune split, give test macro P@5 0.538, so the result does not hinge on the hand-set $0.6/0.2/0.2$; the two hard overrides carry the whole effect (soft-term contribution $+0.000$ against $+0.006$ on the deployment). (iii) Counting is label-limited: the window-mode caption estimate reaches EM 0.10/MAE 2.20 against the anchor-frame caption count, while counts in one window span 4.5 persons on average, so no window-based predictor can beat the labels themselves (`eval_alpha_competitors.py`).

C2 – fusion ablation: The heterogeneous-space concat descriptor reaches $F_1@10 = 0.807\,(\pm 0.024)$: +10.7 pp over the best unimodal index (0.700 ± 0.031; paired bootstrap 95% CI $[+7.9, +13.4]$, $p < 0.0002$) and +2.2 pp over the best fusion baseline RRF (0.785 ± 0.026; 95% CI $[+0.4, +4.1]$, $p = 0.018$, Wilcoxon $p = 0.022$). A single-seed random-Gaussian-projection additive baseline gains +5.4 pp over visual-only ($p = 0.003$) yet stays 5.3 pp below concat ($p = 0.001$); both are significant, the concat-vs-RRF margin least conservative. A deployment gallery-weight sweep gives $F_1@10$ 0.785/0.799/0.774 at $(\alpha_v, \beta_t) = (0.5, 0.5)/(0.7, 0.3)/(0.4, 0.6)$: the RRF tie is a property of that weight, not the family. Multi-seed means for the stochastic same-space projection (five seeds, P@5 0.436 ± 0.0004) and a learned-projection competitor come from the released subset, whose 312k index and per-query export are unreleased (`eval_fusion.py`). *Released-subset scope.* It carries pseudo-tracks, not identity labels, so it cannot test the deployment fusion result and inverts its ordering (concat below visual-only) because pseudo-tracks were formed with visual-only cosine, biasing it upward; it ships as a code-path check (`results/fusion_tables.md`); Table II remains the evidence for C2.

C3 – alert authoring: The 60 held-out utterances exercise all nine rule fields, per-field exact match: event family 0.97, object set 0.95, zone 0.88, threshold τ 0.93, time window 0.92, severity 1.00, cooldown 0.98, side-effects 0.95, cadence 1.00; closed-vocabulary defaults (severity `info`, cadence 1Hz) reach 1.00 when unspecified, so specified-field accuracy is lower. Hardest fields: zone naming (0.88; fuzzy references such as *"the back entrance"* vs. canonical `Zone B`), time window (0.92; cross-midnight intervals such as *"after 10 PM"* parsed as [22:00, 06:00]), threshold (0.93; implicit quantifiers such as *"a few"* missed) and object-set expansion (0.95; category nouns under-expanded, *"vehicle"* → {`car, truck`}); others are event family (*"loiter"* as `zone`) and an implied `snapshot` side-effect. Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt (all 9 fields exact); the remaining 9% need one zone- or threshold-level correction; no structural rewrite was observed.

Released alert benchmark: The release ships 120 author-constructed utterances (60 clean; 60 stress covering ambigu-

ous, conflicting, missing and unsupported requests) with a 30/30 dev/test split; the stress set exercises the clarify/abstain path of `alert_validator.py`. With the deterministic materializer and validator, all 30 clean *test* utterances yield nine exactly correct fields and all 30 stress *test* utterances are clarified, with no false materialization. Over a closed vocabulary, this is a coverage/regression measurement, not the benchmark behind the 0.91 rule-correctness of §VII, which comes from the separate 60-utterance deployment study (Limitations).

Supporting numbers: Substrate metrics: person detection precision/F_1/mAP@0.5 $0.965/0.927/0.873$, tracking MOTA/IDF1 $0.909/0.865$ (mean over 4 cameras), seven-attribute macro F_1 0.882, colour naming CIEDE2000/RGB-Euclid $0.923/0.821$ (1,568 ROIs) and NL field-correctness 0.94; the Z-score detector (Eq. (7)) is in the text.

The 4-camera deployment sustains 2.0 FPS per stream for detection/tracking at 4.2 GB RAM, 7.8 GB VRAM and 68% GPU utilization (91% p95) on the RTX 4070; VLM captioning (Qwen2-VL-2B with fallbacks) runs on keyframes only, outside the 2-FPS budget. End-to-end NL query latency is interactive-only (mean 6.0s, Find 12.0s): per-category parse 1.6–6.4 s, semantic-only lookup 82–137 ms (mixes the 512-d scene index with the 1152-d person index; mean 108.6 ms), total 3.1–12.0 s (LLM parsing, blending, MMR, clip stitching included); FAISS is not the bottleneck. These deployment measurements lack a preserved per-query export; the 109 ms person-index median over N=312,544 descriptors is a separate measurement. `backend/scripts/measure_latency.py` profiles the released subset (CPU, 242 scene and 1,778 person vectors): scene core 37.4ms mean/37.4ms p50/39.7ms p95, person core 17.2ms mean/18.5ms p50/20.4ms p95, 1–8 workers sustaining 26.5–39.5 queries/s at p95 job latency 39.6–265.4ms; these exclude LLM parsing, clip stitching, MongoDB and ingestion and are not 312k-descriptor percentiles.

VIII. Discussion and Conclusion

Limitations. (i) Observation 1 is empirical, assumes steep degradation away from per-category optima and is not guaranteed in general; crowded scenes, occlusion, camera-domain shifts and ambiguous action phrasing degrade both branches, unisolated here. (ii) Concat fusion costs $d_v + d_t$ bytes/descriptor (≈ 1.44 GB for 312k flat); $N > 10^7$ galleries would motivate IVF-PQ over flat IP; MMR (λ=0.3) and recency are unablated. (iii) The 0.91 deployment alert result comes from a held-out study whose per-utterance export was not preserved; the release ships a deterministic keyword/regex materializer routing ambiguous zones, implicit thresholds, cross-midnight windows and unsupported commands to a clarify/abstain validator (30/30 clarifications, no false materializations); the paths are not interchangeable, the released coverage measure is not an open-world accuracy estimate, and no OSNet-only or public-ReID baseline is reported. (iv) *Demographic bias, privacy and consent.* PAR-crossroad-0238 trained on a corpus skewed toward adult, daylight imagery, so accuracy degrades for children, occluded subjects, low-light scenes and under-represented clothing styles; its binary *gender* attribute is a coarse visual proxy, and no demographic-bias evaluation is possible without demographic labels. Live capture was in notified premises, face imagery retained per-camera for at most 8 h. The release ships matching controls—opt-in role/camera/zone access control with audit logging, an 8 h retention prune, a face-blur utility and a gender-off output filter—disabled by default, so the demo is unchanged; they need an operator and a trusted reverse proxy, do not rebuild the shipped index, and the face-blur utility is not wired into live ingestion. Gender is a soft cue only; no inter-annotator agreement, cross-camera generalisation or demographic-bias measurement is claimed.

Conclusion. The three contributions raise macro-mean P@5 from 0.67 to 0.83, gain 10.7-/2.2-pp F_1@10 over unimodal/RRF baselines and reach 91% full-rule alert-authoring correctness on a 4-camera, 312k-descriptor deployment: schema-aware fusion narrows the gap between dense retrieval and a usable conversational surveillance interface. They are schema-decoupled: C1 a category function with finitely many overrides; C2 any pair of frozen encoders; C3 a typed-schema extension per event family.

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